# 7559 | Dynamic contact patch estimation from 3D contact patch measurements inside a moving tyre

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**A B S T R A C T**

Vehicle mobility is inherently related to tyre-terrain interaction. Slip conditions arising from tyre forces generated due to the terrain and vehicles state is a valuable indication of a vehicle's stability. Measurement and analysis of the tyre contact patch is complex, yet abundant information can be derived from contact patch measurements.

Literature indicates that analysis of the contact patch is difficult, and indirect methods are used to determine tyre states under ideal conditions of paved roads. Few researchers attempt to directly analyse the contact patch and extract key tyre states: slip angle, pressure distribution, terrain deformation, contact patch velocity and dynamic rolling radius.

This study uses the T2Cam apparatus, a mechanism that fits into a tyre and provides the capability to attach cameras inside the wheel that remain stationary despite the tyre rotating. The use of T2Cam and an RGBD camera provides the capability of a non-contact method using DIC (Digital Image Correlation) to measure tyre contact patch deformation and relate that to key tyre states. The purpose of this study is to develop algorithms to extract key states from contact patch deformation of the inside of the tyre carcass under varying tyre conditions.

Preliminary findings show tyre deformation of the contact patch is measurable and accurate to be used as an input in an algorithm to calculate slip angle, contact patch velocity and dynamic rolling radius. Overall, determining these factors will enhance the understanding and visualisation of the dynamic contact patch and prediction of vehicle stability on many types of terrain.

*Keywords*
Digital image correlation
Tyre contact patch
Slip angle
Pressure distribution
Dynamic rolling radius
Contact patch velocity
Intelligent tyres

## 1. Introduction

A vital aspect of vehicle dynamics lies in the tyre-terrain interaction. The mechanism through which this interaction takes place is the tyre contact patch and forms a basis on how tyres generate forces on which the vehicle demands. When generating these forces, the tyre deforms in different ways under different conditions: stationary, rolling, cornering, braking, and accelerating. This deformation that the tyre experiences, both in the tyre sidewall and the contact patch, becomes a factor of interest due to its relationship with tyre force generation. Therefore, understanding the tyre contact patch would be extremely valuable in not only confirming the validity of current research but also in investigating and relating the vehicle state to the information derived from the contact patch.

Key tyre states, slip angle and longitudinal slip, are physically developed in the contact patch region and is linked to deformation behaviour in this region, especially due to terrain.

The concept of intelligent tyres is newly formed where it is thought that the combination of embedding sensors into the tyres will inform valuable data and information from them. Intelligent tyres are an important concept to consider when investigating the tyre-terrain interaction.

Literature exists whereby key vehicle states are measured indirectly or contact patch measurements are static. There are limitations in the implementations that leverage the non-contact 3D measurements whereby systems are fixed to external test rigs or move externally with the vehicle. The Vehicle Dynamics Group at the University of Pretoria leverages an apparatus designed by Guthrie (2016) to measure internal deformation of a moving tyre. This system has been the subject of many proceeding works, whereby Kabutz and Els (2024) are able to derive the outside contact patch from the inner deformation 3D measurement.

This work focuses on building upon Kabutz and Els (2024) to dynamically measure the contact patch area and shape, as well as explore the measurement of slip angle and longitudinal slip. Consequently, this work aims to use digital image correlation to measure the tyre deformation directly in a non-contact fashion to provide a custom hardware and real-time capable software infrastructure. The current software pipeline is capable of:

1. Close depth 3D point tracking
2. Tyre contact patch deformation

Ultimately, this research will aim to investigate the use of computer vision to dynamically measure the contact patch from the inside of the tyre. This investigation focuses on specifically finding the shape of the outside contact patch such it can be studied under the influence of varying slip angle.

## 2. Related Work

Currently, there are researchers that statically and dynamically measure the contact patch. The static measure of the contact patch does provide actionable information, however, the dynamic measurement of the contact patch provides valuable information in terms of the deformation of the contact patch under various conditions, such as speed, slip angle and terrain.

Static measurements of the contact patch have been attempted and reveals viability in contact patch measurements with varying change in tyre parameters. Optical methods using cameras placed under glass to measure contact area have already been attempted. (Derafshpour et al., 2019) Additionally, Liang et al. (2021)

https://doi.org/10.56884/ABCD1234 *(ed. note: ISTVS will update your paper's DOI)*
ISBN 978-1-942112-58-7 © 2025 International Society for Terrain-Vehicle Systems.

investigates the pressure distribution estimation using frustration of total internal light reflection (FTIR) methods through a glass plate and validates the contact patch shape under differing conditions. Tyre tread depth measurement using stereovision from the outside of the tyre had been investigated with sub-millimetre accuracies below 0.2mm (Wang et al., 2019). Furthermore, contact area deformation on noise, grip and rolling resistance were investigated (Wang et al., 2021) in which firm correlations were made. Slip angle measurement comparison from sidewall deformation using digital image correlation was investigated and provided positive results (Gao et al., 2024).

This reveals the fact that exploration should take place on experimental setups that are not stationary but move with the vehicle and is able to continuously analyse the tyre contact patch.

## 3. Methodology / System Design

To investigate the deformation of the tyre, a full pipeline must be designed and deployed. Two approaches were explored to measure deformation in the contact patch.

The first approach explores measuring the inner deformation of a tyre by comparing the Red, Blue, Green, Depth (RGBD) image of the deformed tyre to a reference 3D geometry that resembles the tyre in its undeformed state. Furthermore, the rubber-incompressibility assumption was used to project the inner deformation to the outside of the tyre to extract outer contact patch deformation.

The second approach explores the tracking of 3D points on the surface of the inner tyre to extract point-wise displacements and velocities over consecutive RGBD image frames. This method uses the comparison with previous frames instead of a 3D reference geometry, while still providing dense deformation data.

Both approaches capture the deformation behaviour in the tyre but differ in assumptions made, comparisons made to as well as in the outputs yielded. The following sections detail the design and implementation of each approach as well as the experimental setup.

### *3.1. Physical Apparatus and Hardware*

This experimental setup consists of using the T2Cam apparatus design by Guthrie (2016), in Fig. 1, whereby the hardware is upgraded to Intel RealSense D405 Depth Camera as its stereovision solution as well as the Nvidia Jetson Orin Nano for compute hardware.

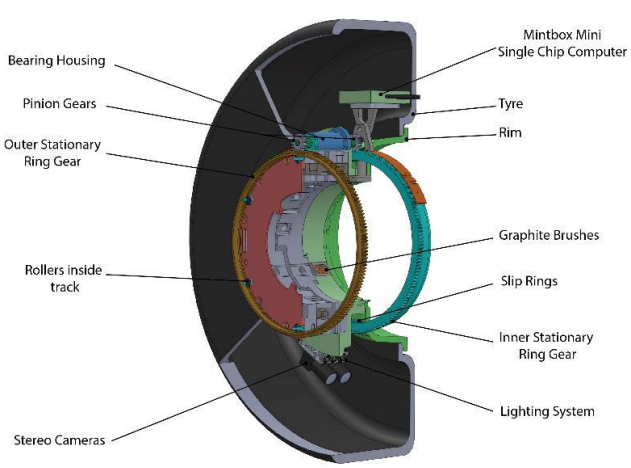


*Fig. 1. T2Cam Apparatus for Camera Stabilisation*

The Intel RealSense D405 delivers synchronised RGBD images at various resolutions and frames per second (fps). The image resolution of 848 x 480 at 30 fps was used to reduce compute cost while retaining high detail at close depths. Additionally, to improve depth accuracy, a speckle pattern on the inside of the tyre was used such that RGB images had high contrast and interest points. The Nvidia Jetson Orin Nano allows for the deployment of real-time capable algorithms by leveraging the edge-computing capabilities of the CUDA cores as well as it supports of many accelerated python libraries.

The mechanism shown in Fig. 1 is designed to mount peripherals inside the tyre and keep it stationary relative to the tyre moving as well as to stabilise Intel RealSense D405 Depth Camera. The stabilisation of this RGBD camera allows for it to be constantly pointing in the direction of the inner surface of the contact patch and thus is the mechanism by which the algorithms were built upon.

### *3.2. Digital Image Correlation and 3D Data Processing*

This work leverages digital image correlation and other computer vision techniques through OpenCV (Bradski, 2000). Additionally, Open3D serves as the 3D data processing library to analyse and measure the contact patch (Zhou et al., 2018) as well as supports Intel RealSense Depth Cameras.

### *3.3. Optical Flow*

Optical flow involves the process of estimating the motion of pixels through two different frames. This algorithm allows for subpixel tracking whereby the displacement in the image plane is estimated. OpenCV implements CUDA accelerated optical flow methods whereby a dense optical flow algorithm was chosen (Bradski, 2000). The Farnebäck dense optical flow method was implemented due to the algorithm modelling local neighbourhoods with polynomial models whereby one can leverage to the continuity of the inner tyre surface geometry (Farnebäck, 2003). Additionally, Farnebäck is more lightweight in terms of computation and more robust to large movements between consecutive frames

### *3.4. 3D Scanning*

Structure from Motion (SfM) methods were used to acquire 3D models, in Fig. 2, depicting the undeformed inner and outer surface geometry of the tyre.

*Fig. 2. Reality Capture Scans of Undeformed Outer Tyre (top) and Inner Tyre (bottom)*

The Reality Capture software pipeline from Epic Games was used and allows for the comparison of the inner deformed tyre surface from the RGBD images with the Reality Capture models in Fig. 2. Therefore, a correspondence between the inner deformed RGBD image and the undeformed 3D inner tyre model was generated. Additionally, the relationship between surface points on the inner undeformed 3D model and the outer undeformed 3D model can be related using the proposed solution from Kabutz and Els (2024). This was done by extending the discretized 2D cross sectional slicing and projection method to 3D ray casting approach using surface normals inherent to the 3D model geometries.

### *3.5. RGBD and Point Cloud Processing*

Each frame captured by the Intel RealSense Depth Camera can be converted into a point cloud and is the data structure that is primarily used to determine deformation in the contact patch. The point cloud, shown in Fig. 3, is an example of the data captured using the T2Cam apparatus and is positioned in such a way that the middle point in the point cloud is central to the static contact patch under vertical load only.

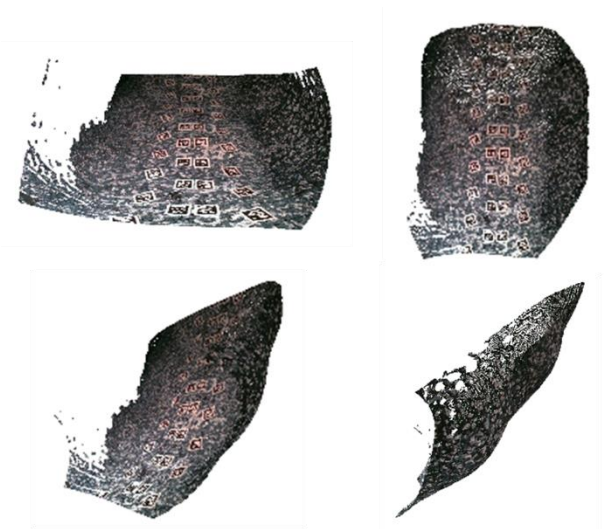

*Fig. 3. RGBD Image from T2Cam Apparatus with point cloud representation with front view (top-left), top view (top-right), diagonal view (bottom-left) and side-on view (bottom-right)*

The Open3D library is leveraged to in terms of its point cloud registration algorithms to correctly position the T2Cam RGBD image relative to the full inner tyre 3D model.

### *3.6. Inner-to-outer surface estimation (Approach 1)*

The first approach to contact patch deformation is implemented through the following process:

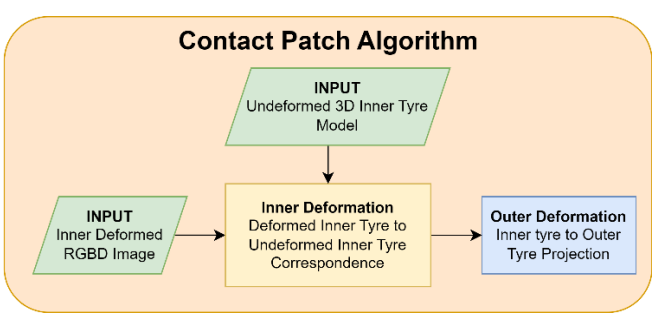


*Fig. 4. Flow Diagram of Summarised Contact Patch Algorithm*

Figure 4 reveals the correspondence between the undeformed inner tyre model and the deformed RGBD image is calculated first. This uses a nearest neighbour search (NNS) approach due to the density of points in both the Reality Capture model and RGBD image. Moreover, the inner undeformed model was used as the source for rough global alignment using April tags (Olson, 2011) and secondly iterative closest point (ICP) registration to refine a more local alignment, such that the NNS approach is valid.

Additionally, the correspondence between the undeformed inner tyre surface and outer tyre surface generated by Reality Capture was processed and represented as the vector distance between these surfaces. Lastly, this vector distance is applied to the inner deformed RGBD image to estimate the outer deformation of each point in that image.

### *3.7. 3D Point-Velocity Tracking (Approach 2)*

The second approach to contact patch deformation is to temporally track points on the inner tyre surface and the given process in summarised in Fig. 5:

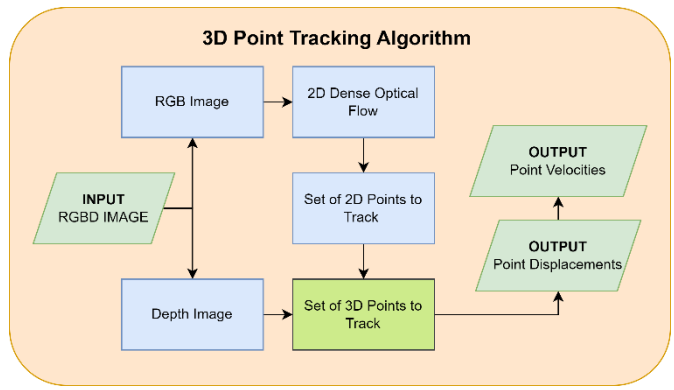


*Fig. 5. Flow Diagram of Summarised 3D Point-Velocity Tracking Algorithm*

This approach uses 2D dense optical flow on all the RGB images from the Intel RealSense Depth Camera, such that image displacements can be tracked with subpixel accuracy for each pixel in the first image. The depth map and camera intrinsics are used to take the 2D point tracking and extend it to a metrically scaled 3D space. This pipeline temporally extends this process to get point displacements and velocities after every frame. This ultimately leads to 3D full-field velocities and displacements, providing motion estimates for each tracked point in the scene.

## 4. Experiments and Results

To evaluate the performance of the two approaches undertaken, test data was captured and applied such that it was able to showcase the visual deformation of the tyre contact patch.

### *4.1. Test Data from Experiments*

The test data consists of a series of 300 RGBD frames as a dataset whereby the Intel RealSense Depth Camera was mounted with the T2Cam apparatus, while **the tyre was rocking back and forth repeatedly**, with the RGBD camera being stationary relative to the tyre. The Trelleborg Agricultural Tyre was used unpressurized and not beaded, such that maximum deformation can be visualised.

*4.2. Test Case: Tyre rocking back and forth with stationary camera relative to tyre: Approach 1*

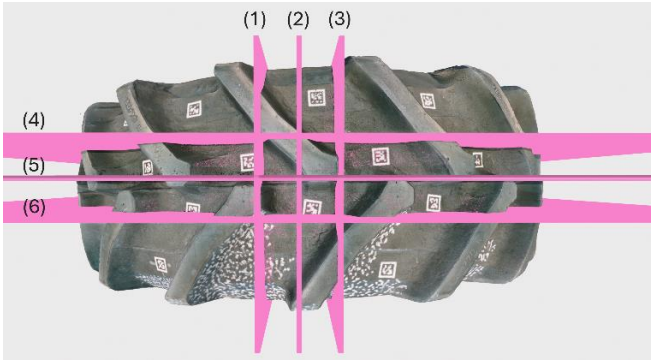


*Fig. 6. Outer Undeformed Model showing the longitudinal and lateral cross-sections to be evaluated (pink) at (1) x = -0.04m, (2) x = 0.0m, (3) x = 0.04m, (4) z = 0.09m, (5) z = 0.13m, (6) z = 0.17m*

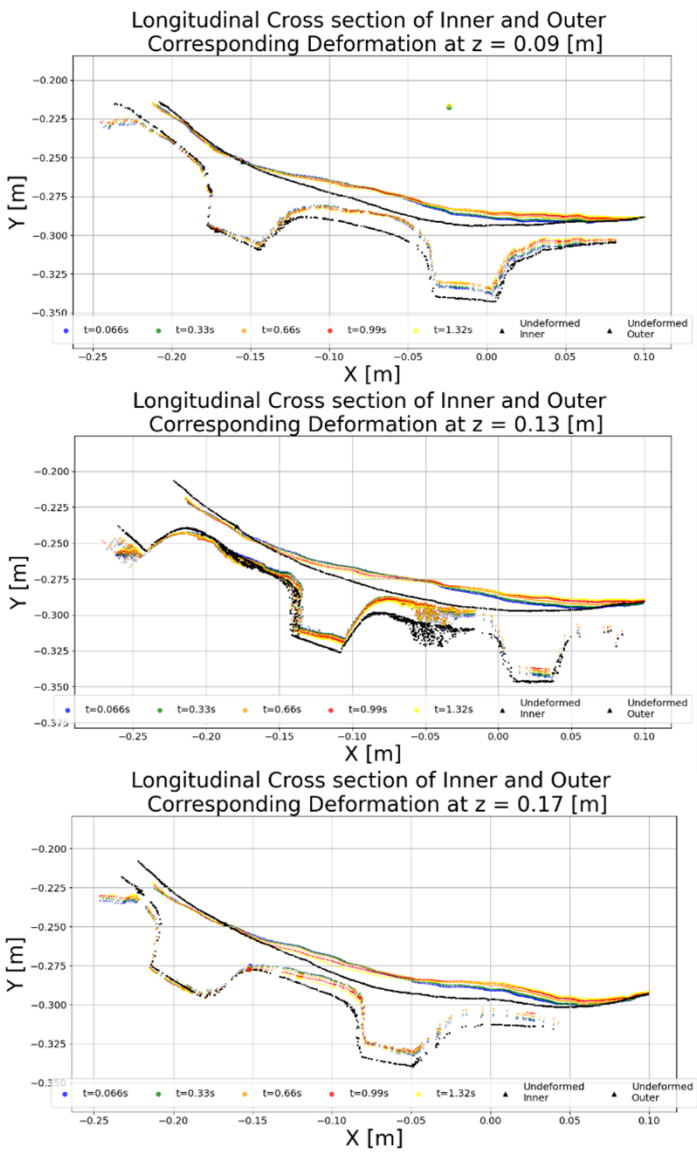


*Fig. 7. Deformation in the Contact Patch over time compared to the undeformed tyre (black) measured with longitudinal cross sections at z = 0.09, 0.13, 0.17m*

In order to evaluate the deformation, cross sections of the tyre were taken in the longitudinal and lateral directions. The pink highlighted planes in Fig. 6 refer to the cross-sections evaluated for:

- Undeformed Inner and Outer 3D Tyre Surface Model
- Five Deformed Inner and Outer Tyre Surface from RGBD Images at t=0.066s intervals

This allows for the comparison between deformed cross sections temporally as well as comparison to the undeformed tyre cross section at each time step.

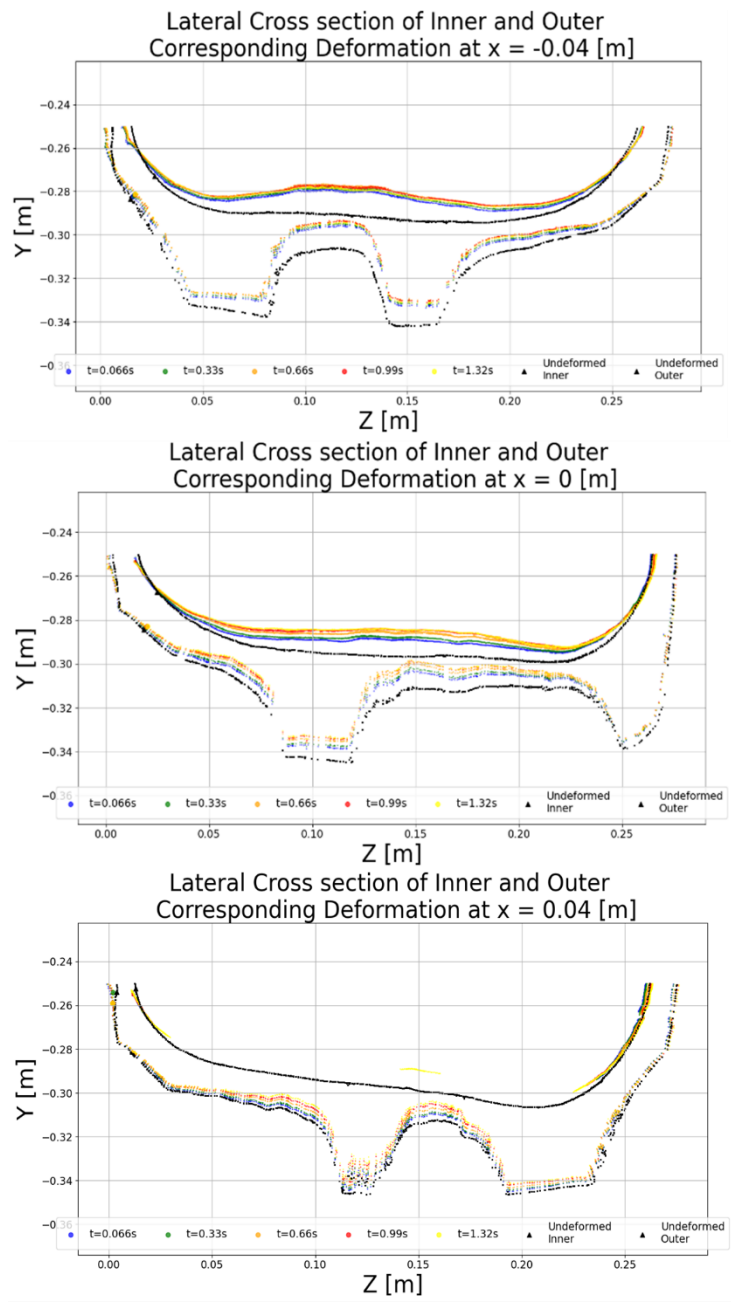


*Fig. 8. Deformation in the Contact Patch over time compared to the undeformed tyre (black) measured with lateral cross sections at z = -0.04, 0, 0.04m*

Figure 7 and 8 illustrates the temporal deformation in relation to the undeformed tyre surface at that specific cross section. The

vertical load on the uninflated tyre, being under its own weight, results in high displacements from the undeformed surface to deformed surfaces. Additionally, the maximum deformation in this case is seen to be centred to the vertical load placed on the tyre. Therefore, as time increases, the tyre rocks forward past t = 0.99s and the vertical load matches the change in which the tyre contacts the ground. Figure 8, emphasis how the deformation increases as the longitudinal z axis approaches the centre of the tyre. Furthermore, Fig. 7 and 8 confirms the non-linearity in the surface deformation as the sidewall has minimal displacement relative to the undeformed surface. Lastly, the tyre lug behaviour is captured on the outer surface estimation, that matches how the vertical load changes across the tyre surface.

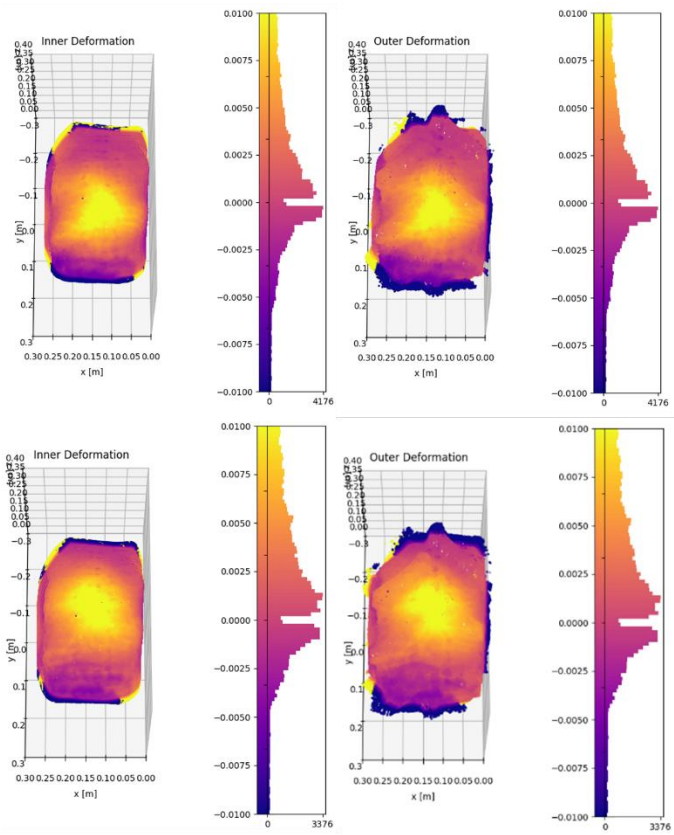


*Fig. 9. 3D Deformation Maps with Histogram of Signed Distances: Inner Deformation at t = 0.066s (top-left), Outer Deformation at t = 0.066s (top-right), Inner Deformation at t = 1.32s (bottom-left), Outer Deformation at t = 1.32s (bottom-right)*

Figure 9 illustrates the point cloud representation of the deformation with accompanying histogram of signed distances. This figure aims to show the extent of the rocking motion applied to the tyre where at t = 0.066s, the highest deformation and, thus, the main contact area, indicated by the yellow region, resides lower relative to the centre. However, while at t = 1.32s, the contact area shifts upwards. Additionally, the histogram of signed distances, reveals a valley at zero, showing the limitations of NNS where point cloud representations of surfaces can yield larger distances than in reality. This confirms that the large tyre lug geometry can be determined from the inside tyre surface only.

Finally, Fig. 7, 8 and 9 visualize the results from different perspectives, but consistently highlight common trends observed in established tyre literature, whereby the contact patch deformation exhibits non-linearity under dynamic and temporal tyre loading conditions.

### *4.3. Test Case 3D Displacement and Velocities of inner tyre surface while rocking back and forth: Approach 2*

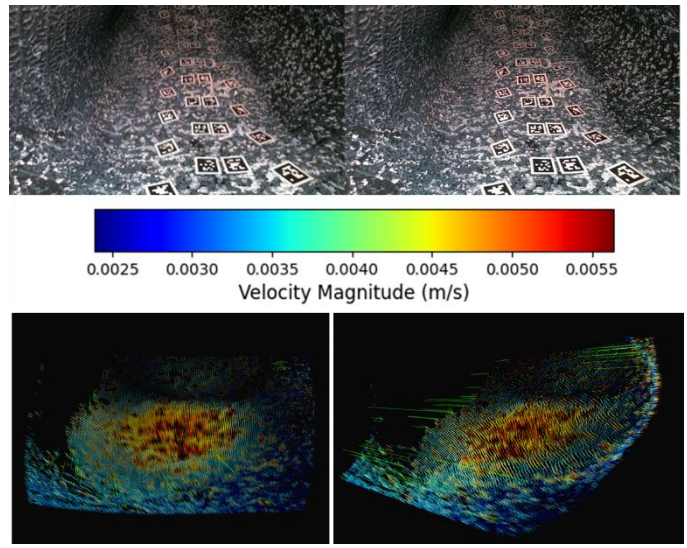


*Fig. 10. 3D Point Tracking across two consecutive frames at t=0.462s to t=0.528s. RGB images of the two consecutive frames (top). Jet colormap corresponding to point velocity magnitudes (middle) and the associated metrically scaled 3D space with velocity vectors at different views (bottom)*

Figure 10 shows the 3D tracking between two consecutive RGBD frames. This approach only tracks the change in surface deformation as the set of initial points to track consist of the deformed surface. Therefore, the point-velocities in red reveals a central shape in the point cloud surface that lines up with the surface normal of the ground on which the tyre is rocked. This approach is robust to small changes in the surface deformation whereby the top-left and top-right frames look indistinguishable to the naked eye. Ultimately, we are able to capture contact patch behaviour from successive frames by which its shape is contrasted well with respect to the surface geometry.

### *4.4. Limitations and Sources of Error*

The results obtained are limited by means of the validation of the deformation of the outer surface due to it being occluded during testing. Additionally, the method by which deformation is measured has many sources of errors, starting with the depth estimation of the RGBD camera, the reconstruction of the undeformed model in Reality Capture, the point cloud registration techniques, as well as the optical flow estimation. All of the above yields a building of errors that compound in such a pipeline used in this work.

## 5. Conclusions and Future Work

Ultimately, the deformation in the contact patch was measured using two approaches in which, multiple outputs can be further derived. The first approach validated the behaviour of the vertical deformation in the contact patch under a rocking motion whereby lateral and longitudinal cross sections temporally deform according to the rocking motion of the tyre. The second approach

validated the behaviour of not only the vertical deformations that occur from the tests but the longitudinal and lateral deformations as the tyre rolls back and forth.

Future work will focus on combining the two approaches in one, such that the point-wise vertical, lateral and longitudinal point-displacements, velocities and deformation can be projected to the outer surface. Additionally, contact patch velocity under of a moving tyre should be derived as well as slip angle which is inherent to lateral and longitudinal velocities in the contact patch of the tyre.

## 6. Acknowledgements

This work is funded by the DSI through the CSIR as part of the South African Government's commitment to promote and support the Collaborative Program in Foundational Digital Capacity Development. Special gratitude is expressed to the Vehicle Dynamics Group at the University of Pretoria.

## 7. Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## 8. Declaration of generative AI in scientific writing

During the preparation of this work the authors used ChatGPT in order to improve paper readability. After using this tool/service, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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